

* light *

- * light : light is a form of energy that enable us to see around us.

We Can Say that :

It is an electromagnetic wave that can travel through medium as well as through vacuum.

- * light - $\left[\begin{array}{l} \text{Particle} \rightarrow \text{photon} \\ \text{Wave} \rightarrow \text{E.M wave} \end{array} \right.$

- Reflection $\rightarrow$ Rectilinear propagation
 $\searrow$ Straight line movement is called Rectilinear propagation

- Speed = 3×10^8 m/s . (Vacuum or Air) .

Maximum Speed $\Rightarrow$ $C = 3 \times 10^8$ m/s

- * Reflection (Reflection of light) :

➤ It is the bouncing back of light rays on striking the surface.

Note :

Good reflection need < opaque smooth

1. light

* Characteristics of Best Reflectors :

- It should have shiny surface.
- It should have polished surface.
- It should have smooth surface.

* Reflection of Light :

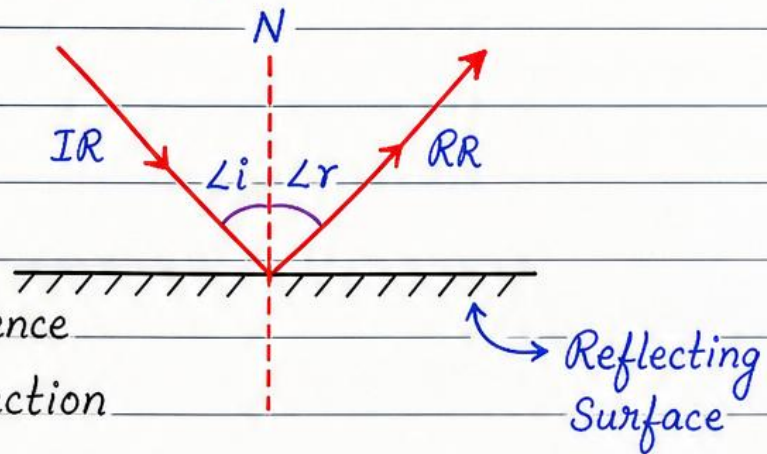
IR = Incident Ray

RR = Reflected Ray

N = Normal

$\angle i$ = Angle of Incidence

$\angle r$ = Angle of Reflection



* Laws of Reflection :

- ✓ Incident ray, reflected ray and normal ~~are~~ all lie on same plane at the point of Incidence.
- Angle of incidence is always equal to Angle of Reflection.

* How do we see leaf (Green Color) ?

➔ All Colours are absorbed except Green, Green is reflected away.

1. light

* Types of Mirrors :

- (i) Plane mirror
- (ii) Spherical mirror
- Concave mirror
- Convex Mirror

* (i) Concave : Concave is that in which deflection occurs through inner surface of mirrors.

* (ii) Convex : It is a type of spherical mirrors in which deflection occurs through outer surface of mirrors.

* Image is associated with deflected rays.



Types of Image :

(i) Real Image :

- Real Image is that which is formed when reflected rays actually meet at certain point.
- It is always inverted.
- It can be obtained on screen.
- It is formed by concave mirrors & Concave to Convex

1. Light

Genius

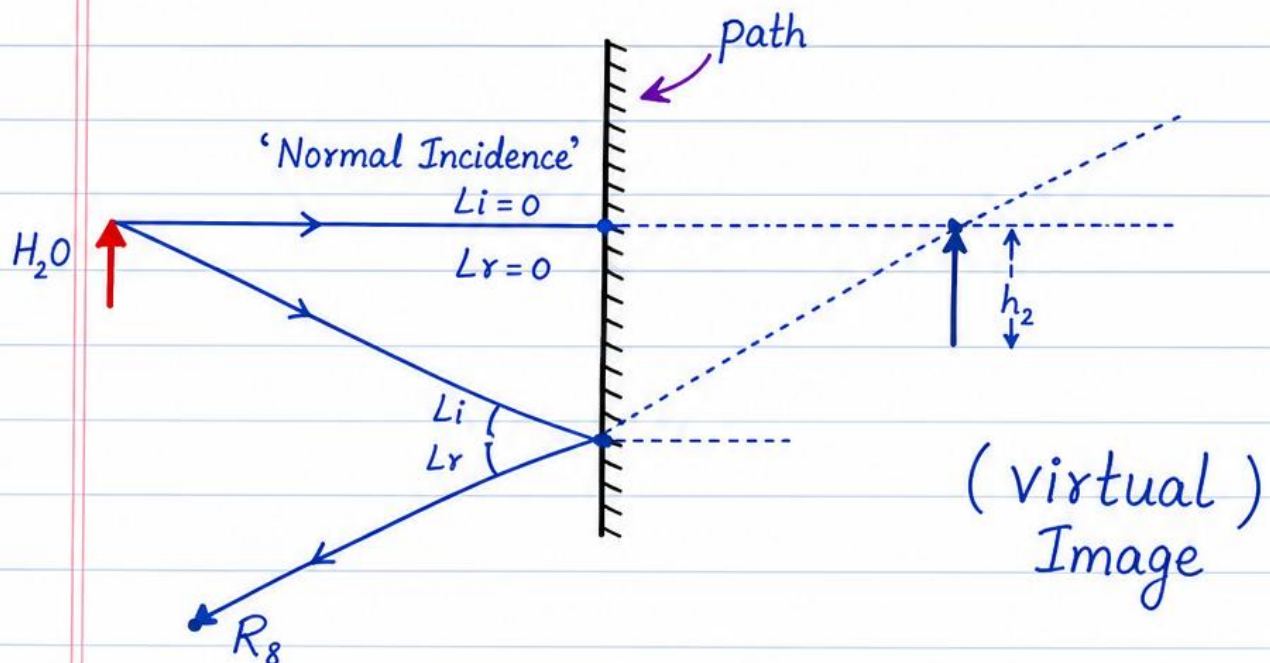
Page No. _____

Date _____

(ff) Virtual image :

- ▶ virtual image is that which is formed when reflected rays do not meet actually but appear to meet at certain point.
- ▶ It is always erect.
- ▶ It Can't be obtained on screen.
- ▶ Screen is not required.
- ▶ It is formed by a Plane Mirrors, Convex mirrors & Concave lens.

* Image Formation by plane Mirrors :



1. Light (Plane Mirror & Spherical Mirrors)

Image formed by plane mirror

* Characteristics of Image:

★ Virtual, Erect

★ Same Size

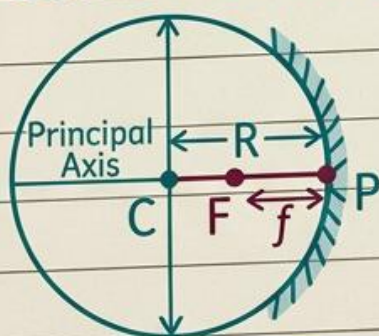
★ Laterally Inverted

* Lateral Inversion:

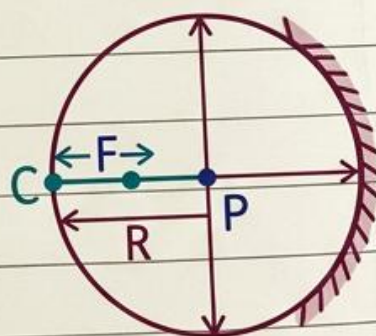
It is the phenomenon when LHS becomes RHS or vice-versa. When an object is exposed to plane mirror.

* Spherical Mirrors

Concave Mirror



Convex Mirror



$CP = R$ (Radius of Curvature)

$FP = f$ (Focal Length)

P = Pole

C = Centre of Curvature

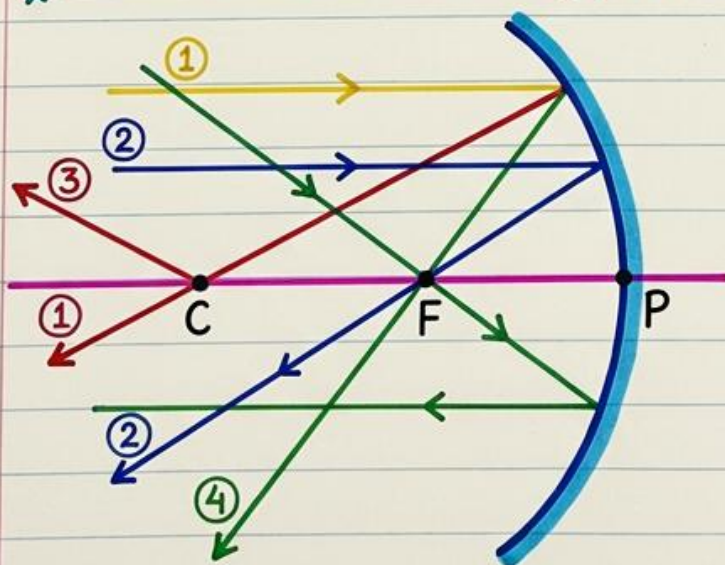
F = Principal Focus

$$R = 2f$$

UNIT: LIGHT

- * **Aperture**: It is a part of spherical mirror from which actually the reflection takes place. The diameter of the spherical mirror.
- * **Centre of Curvature**: It is Centre of the spherical mirror.
- * **Radius of Curvature**: It is the distance between 'C' and 'P'.
- * **Focus**: It is the point where all reflected rays meet or appear to meet.
- * **Pole**: It is the point where p. axis strikes the mirror.

RULES to obtain image from CONCAVE MIRROR



Summary Rules

1. $\parallel$ to P-Axis $\Rightarrow$ F
2. F $\Rightarrow$ $\parallel$ to P-Axis
3. C $\Rightarrow$ Backtrack
4. P $\Rightarrow \angle i = \angle r$

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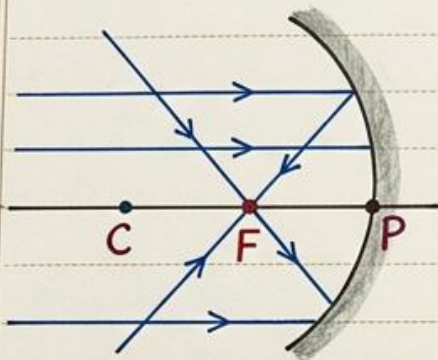


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2nd : The incident ray appears to be coming from 'C' and rebounds back

★ Image Formation : Concave Mirror (Part 1)

1. Object at Infinity

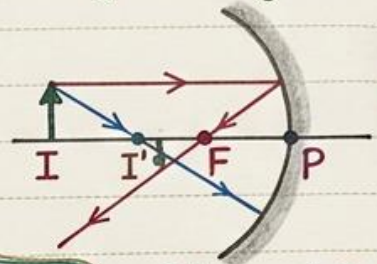


Nature of Image

1. At F
2. Highly diminished
3. Real
4. Inverted

Note : Diminished means very small

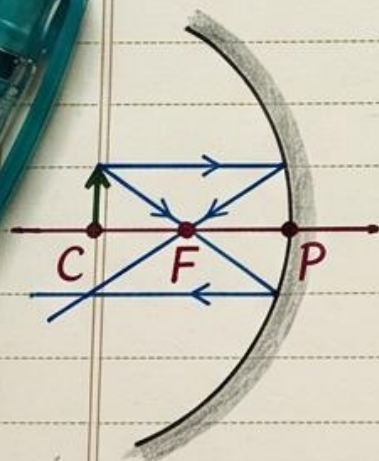
2. Object beyond C



Nature of Image

1. B/w C & F
2. Diminished
3. Real
4. Inverted

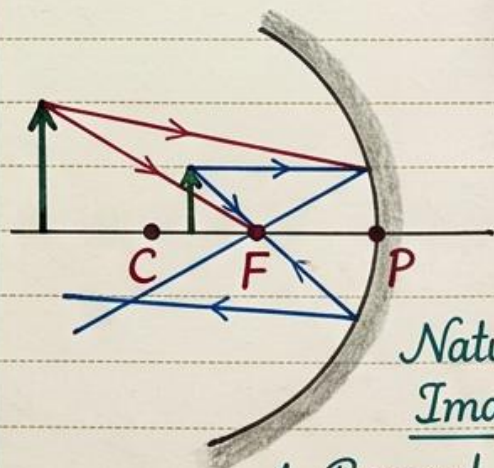
3. Object at C



Nature of Image

1. At C
2. Same Size
3. Real
4. Inverted

4. Object between C & F



Nature of Image :

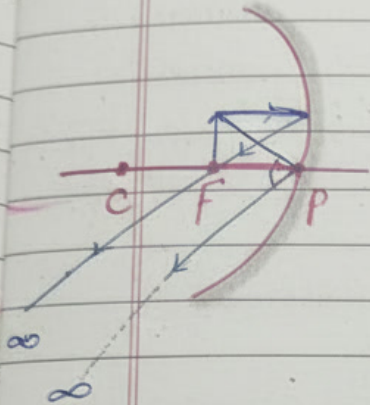
1. Beyond C
2. Enlarged
3. Real
4. Inverted

Image Formation : Concave Mirror (3)

(5) Object at F

Nature of Image:

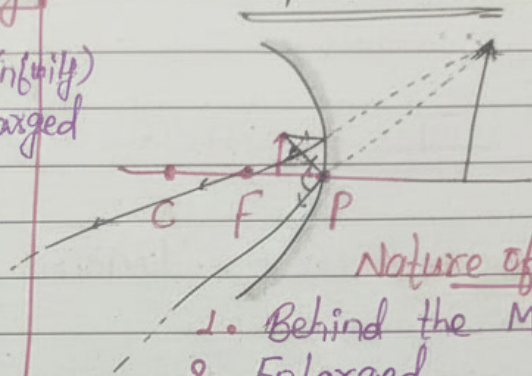
1. At ∞ (Infinity)
2. Highly enlarged
3. Real
4. Inverted



★ (6) Object Between F & P

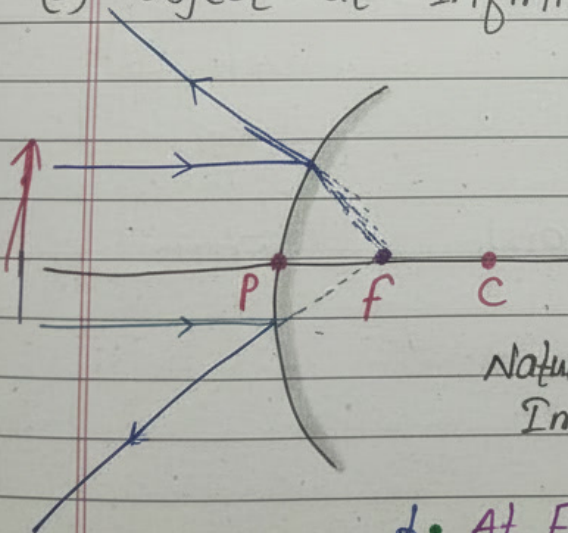
Nature of Image:

1. Behind the Mirror
2. Enlarged
3. virtual (4) Erect



* Image Formation : Convex Mirror

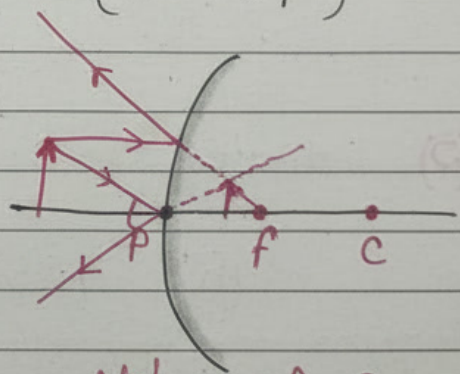
(1) Object at Infinity



Nature of Image:

1. At F
2. Highly diminished

(2) Finite Distance ($\infty \rightarrow P$)



Nature of Image:

1. B/w P & F
2. virtual & erect

Thank You! ♥

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सपने वो हैं जो आपको सोने नहीं देते।



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और जीतने वाले कभी हार नहीं मानते।



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